



Scrum and Agile project management are modern approaches to delivering projects that emphasize flexibility, collaboration, and continuous improvement.



Agile Project Management

Agile is an **iterative and adaptive methodology** for managing projects. Instead of rigid, sequential phases (like Waterfall), Agile delivers work in **short cycles (sprints)** and adapts based on feedback.

Core Principles (Agile Manifesto):

- Individuals and interactions over processes and tools.
- Working software over comprehensive documentation.
- Customer collaboration over contract negotiation.
- Responding to change over following a plan.

Purpose: Enables teams to respond quickly to changing requirements and deliver value continuously.



Scrum Framework

Scrum is a **specific Agile framework** that organizes work into **sprints (2–4 weeks)** with defined roles, events, and artifacts.

Roles:

- **Product Owner:** Owns product vision and backlog.
- **Scrum Master:** Facilitates process and removes blockers.
- **Development Team:** Builds product increments.

Events:

- **Sprint Planning** → Define sprint work.
- **Daily Scrum** → 15-minute team sync.
- **Sprint Review** → Inspect increment with stakeholders.
- **Sprint Retrospective** → Reflect and improve.

Artifacts:



- **Product Backlog** → Ordered list of all work.
- **Sprint Backlog** → Subset of backlog for sprint.
- **Increment** → Completed, potentially shippable product.

Agile vs Scrum

<u>Aspect</u>	<u>Agile</u>	<u>Scrum</u>
Definition	Philosophy/methodology	Specific framework within Agile
Scope	Broad principles	Defined roles, events, artifacts
Flexibility	High, adaptable	Structured but iterative
Cycle	Iterations	Sprints (2–4 weeks)
Focus	Customer collaboration & adaptability	Delivering increments of value

✨ Insight ✨

Think of **Agile** as the **mindset** (values and principles) and **Scrum** as one of the most popular **frameworks** that puts Agile into practice. Agile says “*be flexible and deliver value iteratively*”, while Scrum gives you the **roles, events, and artifacts** to make that happen in real projects.



The foundation of the framework: **Product Owner, Scrum Master, and Development Team**. Each has distinct responsibilities that balance vision, facilitation, and execution.



Product Owner

The person responsible for maximizing the value of the product.

Responsibilities:

- Owns the **product vision** and communicates it clearly.
- Manages and prioritizes the **product backlog**.
- Defines **user stories** and acceptance criteria.
- Acts as the **bridge between stakeholders and the team**.

Contribution: Ensures the team builds features that deliver the most business value.



Scrum Master

A facilitator and coach who ensures Scrum principles are followed.

Responsibilities:

- Organizes **daily stand-ups, sprint reviews, and retrospectives**.
- Removes **impediments** that block team progress.
- Shields the team from external distractions.
- Coaches the team and stakeholders on **Agile practices**.

Contribution: Creates an environment of collaboration, focus, and continuous improvement.



Development Team

A self-organizing, cross-functional group that builds the product increment.

Responsibilities:

- Deliver potentially shippable product increments each sprint.
- Collaborate closely with the Product Owner and Scrum Master.
- Estimate work, commit to sprint goals, and maintain quality.
- Continuously improve processes and technical practices.

Contribution: Transforms backlog items into working solutions that meet customer needs.



Comparison Table



<u>Role</u>	<u>Focus</u>	<u>Key Responsibility</u>	<u>Outcome</u>
Product Owner	Vision & backlog	Prioritize features, define user stories	Deliver business value
Scrum Master	Facilitation & coaching	Remove blockers, enforce Scrum	Smooth team workflow
Development Team	Execution & delivery	Build product increments	Working solutions

✨ Insight ✨

Think of it as a **triangle of success**:

- **Product Owner** defines *what* to build.
- **Scrum Master** ensures *how* it's built follows Agile principles.
- **Development Team** delivers the *actual product*.



In **Scrum**, the framework defines several **artifacts** that provide transparency and alignment for the team. The three core artifacts are the **Product Backlog**, **Sprint Backlog**, and **Increment**. Let's break them down:



Artifacts

Artifacts are **information containers** in Scrum that represent work and value.

Purpose: They ensure transparency, accountability, and alignment between the Scrum Team and stakeholders.



Product Backlog

An **ordered list of everything needed in the product**, maintained by the Product Owner.

Contents: Features, enhancements, bug fixes, technical tasks, research items.

Characteristics:

- Dynamic and evolving.
- Items are prioritized by business value.
- Provides a single source of truth for future work.

Owner: Product Owner.



Sprint Backlog

A **subset of the product backlog** selected for the current sprint.

Contents: User stories, tasks, and goals the team commits to completing within the sprint.

Characteristics:

- Owned by the Development Team.
- Updated daily during the sprint.
- Represents the team's plan for delivering the sprint goal.



Increment

The **sum of all completed backlog items** during a sprint, plus the value of previous increments.



Characteristics:

- Must meet the **Definition of Done (DoD)**.
- Represents a potentially shippable product.
- Provides transparency on progress toward the product goal.

Owner: Development Team ensures quality; Product Owner decides release.



Comparison Table



<u>Artifact</u>	<u>Definition</u>	<u>Owner</u>	<u>Purpose</u>
Product Backlog	Ordered list of all work	Product Owner	Defines product vision & priorities
Sprint Backlog	Subset of product backlog for sprint	Development Team	Guides sprint execution
Increment	Completed work meeting DoD	Development Team	Delivers usable product value

✨ Insight ✨

Think of it like this:

- **Product Backlog** = *Everything we could do.*
- **Sprint Backlog** = *What we will do now.*
- **Increment** = *What we have done and can deliver.*



In **Scrum**, the framework defines **events (ceremonies)** that create rhythm, transparency, and opportunities for inspection and adaptation. The four key events are **Sprint Planning, Daily Scrum, Sprint Review, and Sprint Retrospective**.



Events

Time-boxed meetings that structure the work of the Scrum Team.

Purpose: Provide regular opportunities to plan, inspect, and adapt.

Core Events: Sprint Planning, Daily Scrum, Sprint Review, Sprint Retrospective.



Sprint Planning

A meeting at the start of each sprint where the team decides **what work will be done** and **how it will be achieved**.

Key Activities:

- Product Owner presents prioritized backlog items.
- Development Team estimates and commits to sprint goals.
- Scrum Master facilitates.

Outcome: A **Sprint Backlog** and a clear **Sprint Goal**.



Daily Scrum

A short (15-minute) daily meeting for the Development Team.

Key Activities:

- Each member shares:
 - What they did yesterday.
 - What they will do today.
 - Any blockers.

Outcome: Team alignment and quick problem-solving.



Sprint Review

A meeting at the end of the sprint to **inspect the increment** and gather feedback.

Key Activities:

- Development Team demonstrates completed work.
- Product Owner discusses backlog status.
- Stakeholders provide feedback.

Outcome: Adapted product backlog and stakeholder alignment.



Sprint Retrospective

A meeting after the Sprint Review where the Scrum Team reflects on **process and collaboration**.

Key Activities:

- Identify what went well.
- Discuss what could be improved.
- Agree on actionable improvements for the next sprint.

Outcome: Continuous improvement in team practices.



Comparison Table



<u>Event</u>	<u>Focus</u>	<u>Participants</u>	<u>Outcome</u>
Sprint Planning	Define sprint work	Product Owner, Scrum Master, Dev Team	Sprint Backlog & Goal
Daily Scrum	Team alignment	Development Team	Shared progress & blockers
Sprint Review	Inspect increment	Scrum Team + Stakeholders	Feedback & backlog updates
Sprint Retrospective	Improve process	Scrum Team	Actionable improvements

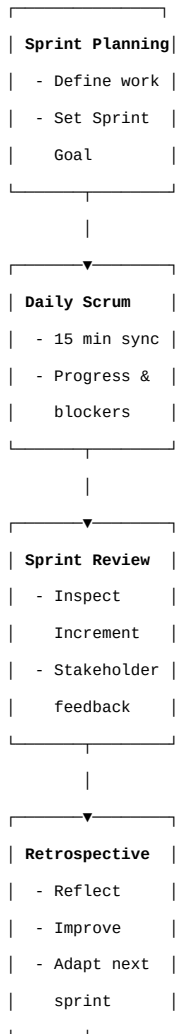
✨ Insight ✨

Think of Scrum events as the **heartbeat of Agile delivery**:

- **Sprint Planning** sets the direction.
- **Daily Scrum** keeps the team aligned.
- **Sprint Review** validates progress with stakeholders.
- **Sprint Retrospective** ensures continuous improvement.



Scrum Events Cycle Diagram



Cycle Flow

- Sprint Planning** → sets the direction.
- Daily Scrum** → keeps the team aligned.
- Sprint Review** → validates progress with stakeholders.
- Sprint Retrospective** → ensures continuous improvement.

✨ **Insight** ✨

This cycle repeats every sprint, creating a **feedback loop** that balances planning, execution, inspection, and adaptation. It's the heartbeat of Scrum.